

1. Kafka Cluster & Topic

```
○ requirements.txt
  WARNING: Ignoring invalid distribution -ip (e:\vscode\lib\site-packag
  es)
  WARNING: Ignoring invalid distribution -ip (e:\vscode\lib\site-packag
  es)
  Requirement already satisfied: kafka-python-ng==2.2.3 in e:\vscode\li
  b\site-packages (from -r requirements.txt (line 1)) (2.2.3)
  Collecting pyspark==3.5.3
    Using cached pyspark-3.5.3.tar.gz (317.3 MB)
    Preparing metadata (setup.py) ... done
  Collecting flask==3.0.0
    Downloading flask-3.0.0-py3-none-any.whl (99 kB)
    99.7/99.7 KB 272.8 kB/s eta 0:00:00
  Collecting pandas==2.2.0
    Downloading pandas-2.2.0-cp39-cp39-win_amd64.whl (11.6 MB)
    5.7/11.6 MB 994.7 kB/s eta 0:00:06
```

```
Desktop\\waike\\Data engineer\\exam\\docker-compose.yml: `version` is
obsolete"
[+] Running 3/3
  ✓ Network exam_default    Created          0.1s
  ✓ Container zookeeper     Started          0.7s
  ✓ Container kafka1        Started          1.0s
○ PS C:\Users\guan\Desktop\waike\Data engineer\exam>
PS C:\Users\guan\Desktop\waike\Data engineer\exam> # 等待 30 秒让 Kaf
ka 启动
PS C:\Users\guan\Desktop\waike\Data engineer\exam> Start-Sleep -Secon
ds 30
PS C:\Users\guan\Desktop\waike\Data engineer\exam>
PS C:\Users\guan\Desktop\waike\Data engineer\exam> # 创建 Topic
PS C:\Users\guan\Desktop\waike\Data engineer\exam> python create_topi
c.py
  ✎ 正在尝试连接 Kafka 并创建 Topic...
  ⚡ 正在发送创建请求...
  ✓ 成功! Topic 'sensor-events' 已创建。
○ PS C:\Users\guan\Desktop\waike\Data engineer\exam> []
```

2. Start Data Producer

Start Sending Data

```
r.py
🚀 开始发送 100 条模拟数据...
🎉 发送完成!
PS C:\Users\guan\Desktop\waike\Data engineer\exam>
```

3. Start Spark Streaming Pipeline

```
最近10条数据统计:
传感器类型: {'humidity': 4, 'pressure': 4, 'temperature': 2}
值范围: 22.43 ~ 94.58
平均值: 57.96
⚠ 发现 3 条异常数据!

最近10条数据统计:
传感器类型: {'humidity': 6, 'temperature': 3, 'pressure': 1}
值范围: 21.58 ~ 89.62
平均值: 53.61
⚠ 发现 3 条异常数据!

最近10条数据统计:
传感器类型: {'temperature': 4, 'pressure': 3, 'humidity': 3}
值范围: 22.17 ~ 98.03
平均值: 62.36
⚠ 发现 5 条异常数据!
```

4. Start Flask API

```
Desktop\waikē\Data engineer\exam
PS C:\Users\guan\Desktop\waikē\Data engineer\exam> python src/api/app
.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production
deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (windowsapi)
* Debugger is active!
* Debugger PIN: 355-730-267
127.0.0.1 - - [19/May/2026 10:03:24] "GET /api/v1/health HTTP/1.1" 200 -
127.0.0.1 - - [19/May/2026 10:03:36] "GET /api/v1/health HTTP/1.1" 200 -
```

5. Test API

```
PS C:\Users\guan\Desktop\waike\Data engineer\exam>
PS C:\Users\guan\Desktop\waike\Data engineer\exam> curl http://localhost:5000/api/v1/health
```

```
StatusCode      : 200
StatusDescription : OK
Content         : {
                  "status": "ok"
                }

RawContent      : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 21
                  Content-Type: application/json
                  Date: Tue, 19 May 2026 02:03:24 GMT
                  Server: Werkzeug/3.1.8 Python/3.9.13

                  {
                    "status": "ok"
                  }

Forms           : {}
Headers         : {[Connection, close], [Content-Length, 21], [Content-Type, application/json], [Date, Tue, 19 May 2026 02:03:24 GMT]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 21
```

← ↻ ⓘ localhost:5000/api/v1/health

优质打印 ☐

```
{
  "status": "ok"
}
```

Send Sensor Data

```
PS C:\Users\guan\Desktop\waike\Data engineer\exam>
PS C:\Users\guan\Desktop\waike\Data engineer\exam> $body = '{"sensor":
:"temperature", "value":25, "timestamp":1234567890}'
PS C:\Users\guan\Desktop\waike\Data engineer\exam> Invoke-WebRequest
-Uri http://localhost:5000/api/v1/readings -Method POST -Body $body -
ContentType "application/json" | Select-Object -ExpandProperty Conten
t
{
  "message": "accepted"
}
```